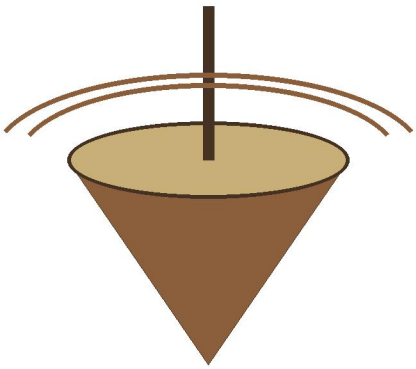
Spinning Top





What to do:

Push a short pencil through the middle of a paper plate.

Spin the pencil tip on a table and let go.

What happens:

The plate stays standing while it spins fast.

As it slows down, it wobbles and falls over.

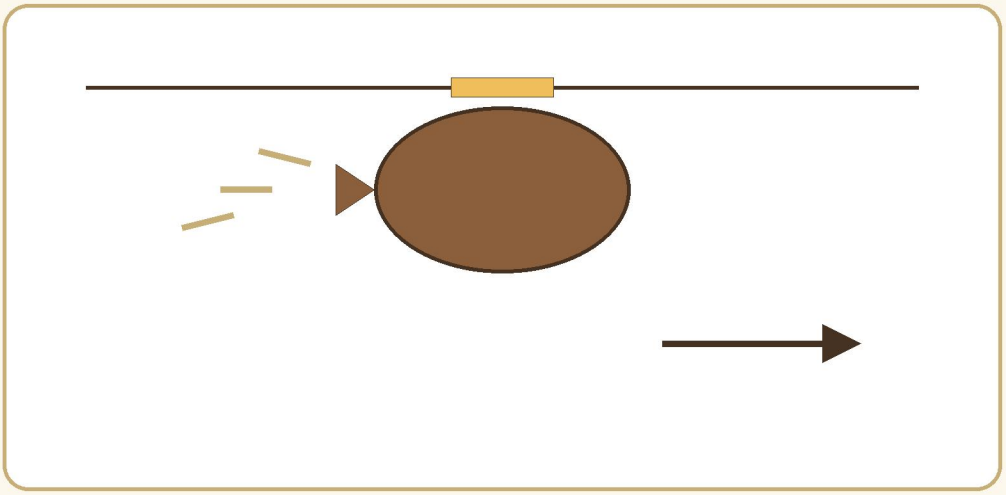
The big idea: rotational inertia

Spinning things like to keep spinning the same way.

The faster it spins, the steadier it stands!

Balloon Rocket





What to do:

Thread a straw onto a long string and tie it tight across a room.
Tape a blown-up balloon to the straw, then let the neck go.

What happens:

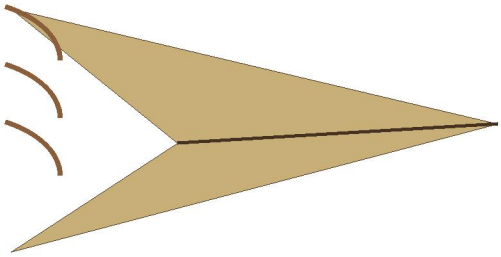
Air rushes out the back and the balloon zooms forward along the string. Whoosh!

The big idea: Newton's third law

Every push has an equal push back the other way.
Air pushes back, so the rocket goes forward.

Paper Plane





What to do:

Fold a paper plane with a pointy nose and wide wings.
Throw it gently, then try bending the wing tips up a little.

What happens:

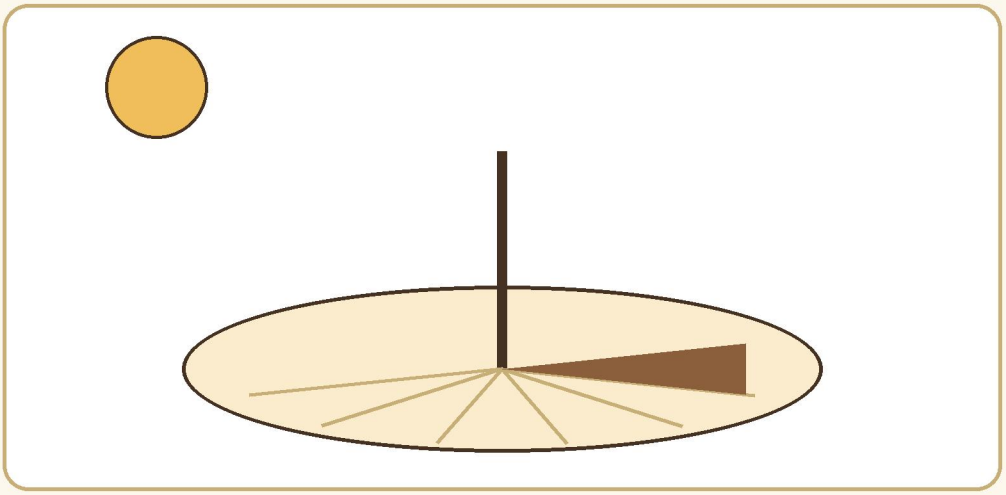
It glides through the air instead of dropping straight down.
Bent wing tips make it climb, turn or loop.

The big idea: aerodynamics

Air flowing over and under the wings holds the plane up.
Changing the wing shape changes how the air pushes it.

Shadow Clock





What to do:

Stand a stick in the ground or in playdough on a plate outside.
Mark the tip of its shadow with a pebble, again and again.

What happens:

The shadow slowly moves and changes length during the day.
Your pebbles make a curve, like a clock!

The big idea: Earth's rotation

The Sun isn't moving across the sky, we are!
Earth spins, so the Sun's light hits the stick from new angles.